

12.367

EE25BTECH11042 - Nipun Dasari

Question:

The eigen-values of a real symmetric matrix are always

- a) positive
- b) imaginary
- c) real
- d) complex conjugate pairs

Solution: Let $\mathbf{A}$ be a real symmetric matrix. This means all its entries are real numbers and $\mathbf{A} = \mathbf{A}^\top$.

Let λ be an eigenvalue of $\mathbf{A}$ and $\mathbf{v}$ be its corresponding eigenvector. Note that λ and the components of $\mathbf{v}$ could be complex. The eigenvalue equation is:

$$\mathbf{A}\mathbf{v} = \lambda\mathbf{v} \quad (4.1)$$

If we take the complex conjugate of this entire equation, we get:

$$(\mathbf{A}\mathbf{v})^c = (\lambda\mathbf{v})^c \quad (4.2)$$

$$\mathbf{A}^c\mathbf{v}^c = \lambda^c\mathbf{v}^c \quad (4.3)$$

Since $\mathbf{A}$ is a real matrix, its complex conjugate is itself ($\mathbf{A}^c = \mathbf{A}$). So, the equation becomes:

$$\mathbf{A}\mathbf{v}^c = \lambda^c\mathbf{v}^c \quad (4.4)$$

Now, left-multiply the original eigenvalue equation by $(\mathbf{v}^c)^\top$:

$$(\mathbf{v}^c)^\top \mathbf{A}\mathbf{v} = (\mathbf{v}^c)^\top (\lambda\mathbf{v}) = \lambda((\mathbf{v}^c)^\top \mathbf{v}) \quad (4.5)$$

Next, take the transpose of both sides of equation (4.4):

$$(\mathbf{A}\mathbf{v}^c)^\top = (\lambda^c\mathbf{v}^c)^\top \quad (4.6)$$

$$(\mathbf{v}^c)^\top \mathbf{A}^\top = \lambda^c(\mathbf{v}^c)^\top \quad (4.7)$$

Since $\mathbf{A}$ is a symmetric matrix, $\mathbf{A}^\top = \mathbf{A}$. The equation is:

$$(\mathbf{v}^c)^\top \mathbf{A} = \lambda^c(\mathbf{v}^c)^\top \quad (4.8)$$

Now, right-multiply this equation by $\mathbf{v}$:

$$(\mathbf{v}^c)^\top \mathbf{A}\mathbf{v} = \lambda^c((\mathbf{v}^c)^\top \mathbf{v}) \quad (4.9)$$

By comparing equations (4.5) and (4.9), we see that:

$$\lambda((\mathbf{v}^c)^\top \mathbf{v}) = \lambda^c((\mathbf{v}^c)^\top \mathbf{v}) \quad (4.10)$$

$$(\lambda - \lambda^c)((\mathbf{v}^c)^\top \mathbf{v}) = 0 \quad (4.11)$$

The term $(\mathbf{v}^c)^\top \mathbf{v}$ is the dot product of the vector $\mathbf{v}^c$ with $\mathbf{v}$, which is the squared norm of $\mathbf{v}$.

$$(\mathbf{v}^c)^\top \mathbf{v} = \sum_i v_i^c v_i = \sum_i |v_i|^2 = \|\mathbf{v}\|^2 \quad (4.12)$$

Since $\mathbf{v}$ is an eigenvector, it cannot be the zero vector, so its norm $\|\mathbf{v}\|^2$ must be a positive real number.

Given that $\|\mathbf{v}\|^2 \neq 0$, the only way for the equation $(\lambda - \lambda^c) \|\mathbf{v}\|^2 = 0$ to hold is if:

$$\lambda - \lambda^c = 0 \implies \lambda = \lambda^c \quad (4.13)$$

A complex number is equal to its own conjugate only if its imaginary part is zero. Therefore, λ must be a real number.